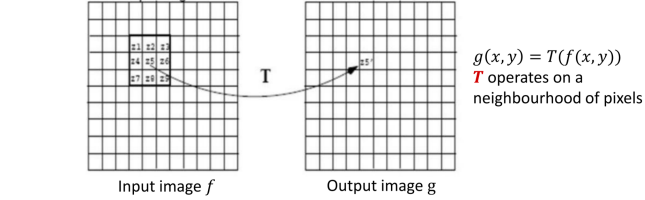
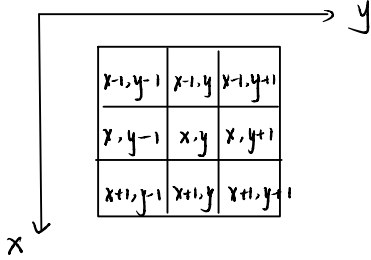


Spatial Filtering

- A spatial filter consists of (a) a neighborhood (e.g., a 3 × 3 neighborhood), and (b) a predefined operation T



- If T is a linear function, then the filter is called linear spatial filter. Otherwise, the filter is called non-linear spatial filter.



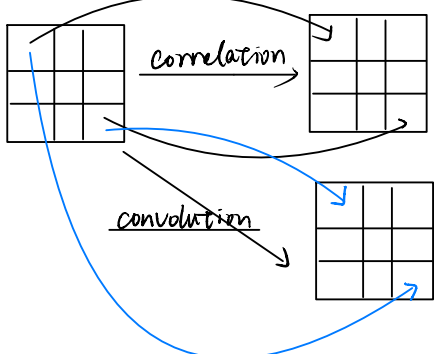
In General, linear spatial filtering of an image of size M × N with a filter of size m × n is given by the expression:

g(x,y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s,t) f(x+s,y+t)

Spatial Correlation

Convolution

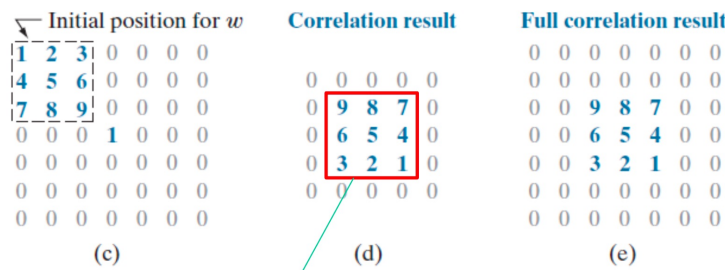
g(x,y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s,t) f(x-s,y-t)



rotate 180°
or "flip" by diagonal.

Zero-padding add 0 on margin

1D convolution \sum_{s=-a}^a w(s) f(x-s)



Property	Convolution	Correlation
Commutative	f * g = g * f	—
Associative	f * (g * h) = (f * g) * h	—
Distributive	f * (g + h) = (f * g) + (f * h)	f \star (g + h) = (f \star g) + (f \star h)

Seperable if w(x,y) = w(x) · w(y)

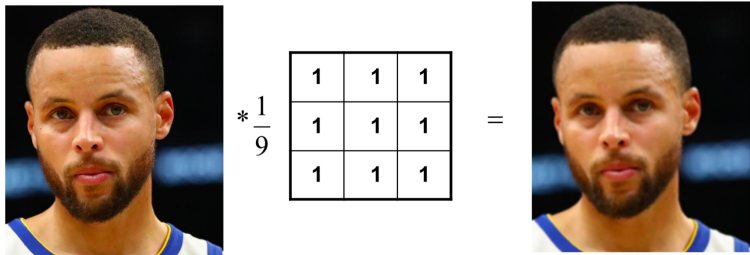
By Associative law of convolution: w * f = (w * w) * f

if 2D convolution: O(MNmn) = w * (w * f)

if 2x1D convolution: O(MN(m+n))

⇒ Much more efficient

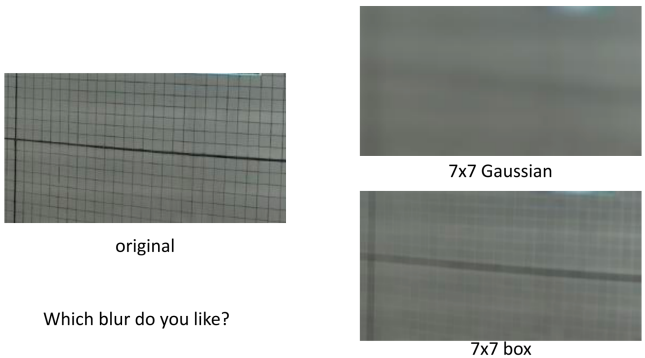
Image averaging R = \frac{1}{9} \sum_{i=1}^9 z_i



Gaussian filter

kernel value Sampled from 2D Gaussian.

w(s,t) = \frac{1}{2\pi\sigma^2} e^{-\frac{s^2+t^2}{2\sigma^2}}



Non-Linear Filter

Order Statistic Filter : Max/Min/Median...

Median Filter

- Smooth images
- Very popular non-linear filter
- Find the median of the window
- Preserve edges
- Remove impulse noise, avoids excessive smoothing

High freq
Lowpass : low freq

Highpass Spatial Filters ⇒ let high freq pass.
low freq removed.
Highlight intensity transition & details
i.e. changes

How to achieve : differentiation

First-order derivative \frac{\partial f}{\partial x} = (f(x+1) - f(x)) / 1 = f(x+1) - f(x)
or = (f(x+1) + f(x-1) - 2f(x)) / 2 ⇒ "0" at constant value.
nonzero at start or middle of ramp

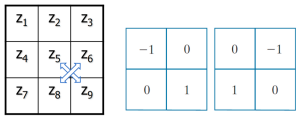
Second : \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} (f(x+1) - f(x))
= f(x+2) - f(x+1) - f(x+1) + f(x)
= f(x+2) + f(x) - 2f(x+1)
or = f(x+1) + f(x-1) - 2f(x) ⇒ Nonzero ONLY at start of ramp
"0" otherwise

First Derivative Filter (Image gradient),

|∇f| = M(x,y) = \sqrt{g_x^2 + g_y^2} \approx |g_x| + |g_y|.
∇f = [\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}]
M(x,y) \approx |g_x| + |g_y|
= |78 - 75| + |78 - 75|

- Gradient for Image Sharpening:

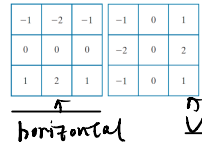
- Roberts Cross-gradient Operators:
M(x,y) \approx |z_9 - z_5| + |z_8 - z_6|



所以边缘算子:
• z5 = z1 和 z4 - z3 → 水平梯度分量 (简单, 但只对垂直边缘敏感).
• z6 = z2 和 z5 - z4 → 垂直梯度分量 (Roberts 算子, 对斜边缘敏感).

- Sobel Operators:

M(x,y) \approx |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)|
+ |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|



Why "0": We first do a Smoothing

When using derivative filters, it is critical to blur first!

if f: ~~~~~ rather than _____
derivative ~~~~~

- Laplacian Operator:

- The simplest second-order isotropic derivative operator is the Laplacian defined as:

\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}
\frac{\partial^2 f}{\partial x^2} = f(x+1,y) + f(x-1,y) - 2f(x,y)
\frac{\partial^2 f}{\partial y^2} = f(x,y+1) + f(x,y-1) - 2f(x,y)

\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)

Highlight transition & edges

Sharpening image

① get Edge map e(x,y) = f(x,y) * h_laplacian

② g(x,y) = f(x,y) + c * e(x,y)

"How much to magnify?"
"magnifying factor"

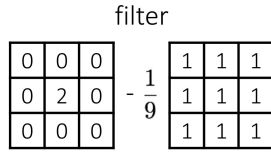
Unsharp masking

g_{sharp} = f + \gamma (f - f_{blur})

"edge" = origin - average

⇒ Highlight the transition part

when \gamma = 1.



do nothing for flat areas
Emphasize intensity peaks

- Unsharp Masking

- Blur the original image
- Subtract the blurred image from the original (the resulting difference is called the mask)
- Add the mask to the original

g_{mask}(x,y) = f(x,y) - \bar{f}(x,y)

g(x,y) = f(x,y) + k g_{mask}(x,y)

k = 1, unsharp masking; k > 1, highboost filtering; 0 ≤ k < 1 reduces the contribution of unsharp mask

k = 1. add back edge 1 time.
k > 1 add back edge k times.
k < 1 add back edge with ratio k

Sharpening – highlight the transitions in intensity by differentiation



Smoothing – blur the transitions by averaging